6(a)(i)	The battery has internal resistance, so when current flow there is a potential drop across internal resistance. Hence terminal pd/voltmeter reading is less than the emf.	[1]
6(a)(ii)	Given: $R_X = 4\ R_Y$ Deduce $V_Y + V_X = 6.0$ V {Kirchhoff's Voltage Law} Since current is the same for both, $V_Y + 4V_Y = 6.0$ V $V_Y = 1.2$ V	 [1] [1]

Source: J2000 P3 Q4

6(b)(i)	Equivalent capacitance of two 50 μF capacitors in series $= (50 \times 50)/ [50 + 50] = 25\ \mu F$ Total capacitance of two 25 μF capacitors in parallel $= 25\ \mu F + 25\ \mu F = 50\ \mu F$	[1] [1]
6(b)(ii)1.	Energy stored in the combination of 4 capacitors when pd reaches 540 V $= ½\ CV^2$ $= ½ \times$ (Ans in (i)) $\times 540^2$ $=7.29$ J Energy dissipated in one flash of light $= 0.63 \times 7.29 = 4.6$ J	 [1,ecf] [1,ecf]
6(b)(ii)2.	Total energy remaining in all 4 capacitors after flash = 37% of 7.29 J = 2.69 J $½\ CV_f^2 = 2.69$ J, where C =Ans in (i) $= 50\ \mu F$ final pd across combination, $V_f = 328$ V Since 2 equal capacitors in series are of equal pd as their charge is the same, pd across each capacitor $= ½ \times V_f = ½ \times 328 = 164$ V	 [1,ecf] [1,ecf]